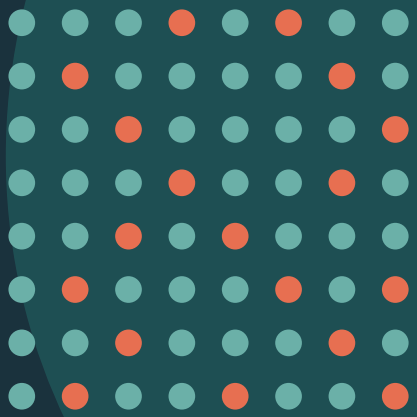
INTERIM PROJECT REPORT



Predicting Customer Churn at AlphaCom

Finding the customers most likely to leave, and why

Business problem | Exploratory analysis | Data preparation | Baseline model

PREPARED BY

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1. Understanding the Business Problem

1.1 Background

AlphaCom is a telecommunications provider with a wide product range and prices that compare well with its competitors. Even so, more customers have been leaving lately. Each departure removes the monthly revenue the customer was paying, and in a crowded market most of them end up with a rival. The retention campaigns used so far treat customers much alike. They do not separate the customers who are about to leave from those who are settled, and they ignore the fact that people leave for different reasons: a high bill, a short contract, little support, an awkward way of paying. The company is reacting to churn after it happens instead of preventing it.

1.2 Problem statement

This is a binary classification problem. The target is **Churn**, with *Yes* (the customer left) as the class of interest.

1.3 Why the study is needed

In the data supplied, 3,394 of 12,028 customers (28.22%) have churned. Churned customers paid more than those who stayed, an average of $73.47 a month against $60.81, so the revenue lost is larger than the head count suggests: they account for 32.23% of all monthly billing in the file. Their combined bills come to about $243,333 a month, or $2.92 million over a year if those customers had stayed on their current plans. Winning a new customer normally costs more than keeping an existing one, so even a modest improvement in retention protects revenue directly.

1.4 Business opportunity

- **Early warning.** A risk score for every customer lets the retention team contact the likeliest leavers first instead of mailing the whole base.
- **Matching the offer to the reason.** If short contracts or manual payment methods drive churn, the offer can be a contract upgrade or an autopay incentive rather than a blanket discount.
- **Better use of budget.** Money for discounts and calls goes to the customers where it changes the outcome, and not to those who would have stayed anyway.

1.5 Objectives

Objective	Question it answers	Where in this report
Understand the customer base	Who are the customers, and how do churners differ from stayers?	Section 2
Prepare reliable data	Which problems does the file contain and how are they fixed without leaking information?	Section 3
Set a reference model	How well can a simple, readable model predict churn, and on which metric should it be judged?	Section 4
Identify the drivers	Which factors raise or lower the chance of leaving?	Sections 2 and 4

1.6 Scope of this report

This report covers the business problem, exploratory analysis, data preprocessing and a baseline logistic regression model. Advanced models, tuning and the final comparison belong to the final submission, and the baseline reported here is the benchmark they must beat. All code and full output are in the supporting notebook.

2. Exploratory Data Analysis

2.1 Data collection and background

The data is a single customer extract supplied by AlphaCom (*customer_churn.csv*). It holds 12,055 records and 20 columns. Each row describes one customer: a few profile details, the services held, the contract and payment arrangements, the months with the company, the amounts billed, and whether the customer has left. There is no customer identifier, so rows cannot be matched back to individuals. All amounts are in US dollars. The 20 columns fall into 3 numeric columns, 1 binary column, 15 categorical columns and the target.

Table 1. *Data dictionary (short form)*

Column	Type	Meaning
gender	Categorical	Male or female
SeniorCitizen	Binary (0/1)	1 if the customer is a senior citizen
Partner	Categorical	Whether the customer has a partner
Dependents	Categorical	Whether the customer has dependents
tenure	Numeric	Months the customer has been with AlphaCom
PhoneService	Categorical	Has a telephone service
MultipleLines	Categorical	Has more than one phone line
InternetService	Categorical	DSL, fiber optic or none
OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport	Categorical	Subscribes to each protection or support add-on
StreamingTV, StreamingMovies	Categorical	Subscribes to each streaming add-on
Contract	Categorical	Month-to-month, one year or two year
PaperlessBilling	Categorical	Receives bills online only
PaymentMethod	Categorical	Electronic check, mailed check, bank transfer or credit card (last two automatic)
MonthlyCharges	Numeric (USD)	Amount billed each month
TotalCharges	Numeric (USD)	Amount billed over the whole tenure
Churn	Target	Whether the customer has left (Yes / No)

2.2 Data overview

The file was read in as it came. Before any chart could be drawn the following problems had to be dealt with; they are listed in Table 2 and treated properly in Section 3.

Table 2. *Data quality problems found in the raw file*

Check	Column	Count	Planned treatment
Churn written in different forms (case / spaces)	Churn	8	Standardise to Yes / No
PaymentMethod written in different forms	PaymentMethod	20	Standardise to 4 methods
Charges stored as text with a currency symbol	MonthlyCharges, TotalCharges	2	Strip symbol, convert to number
Placeholder text '$nan' / '£nan' in place of a value	MonthlyCharges	299	Treat as missing
Placeholder text '$nan' / '£nan' in place of a value	TotalCharges	1,205	Treat as missing
Missing values	tenure	604	Impute (Section 3)
Negative tenure (impossible)	tenure	127	Treat as missing
Exact duplicate rows	all columns	2	Remove

For the exploration, labels were standardised, the currency symbols removed and the impossible values (negative tenure and negative totals) blanked. No rows were dropped and nothing was filled in, so the charts below use all 12,055 records and ignore missing cells chart by chart. The numeric columns, after that light clean-up, look like this:

Table 3. Summary statistics of the numeric columns

Variable	Count	Mean	SD	Min	25%	Median	75%	Max	Skew
tenure	11,324	31.60	24.93	0.00	7.00	28.00	54.00	74.00	0.25
MonthlyCharges	11,754	64.37	30.33	15.29	30.31	71.35	89.38	121.67	-0.21
TotalCharges	10,703	2,323.93	2,276.32	0.00	419.80	1,364.30	4,008.55	9,039.92	0.89

2.3 Univariate analysis

2.3.1 Target variable

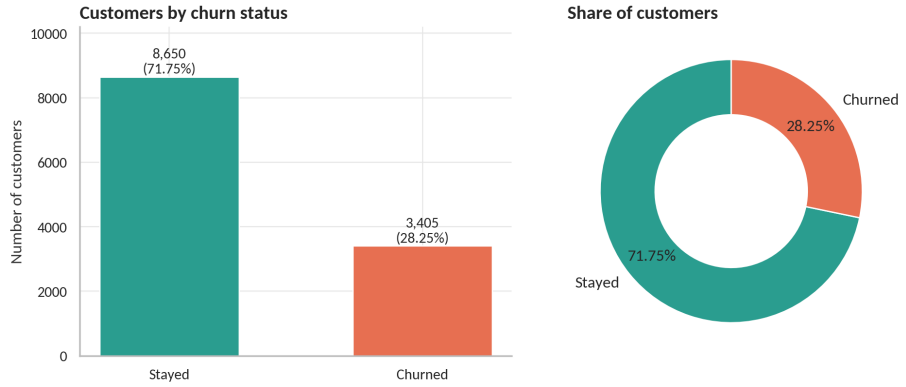


Figure 1. Churn status of all customers.

28.25% of the customers have churned (3,405 of 12,055), a ratio of about 2.5 stayers to every leaver. The target is imbalanced but not extremely so, which matters for how the model is trained and scored (Sections 3.9 and 4).

2.3.2 Numeric variables

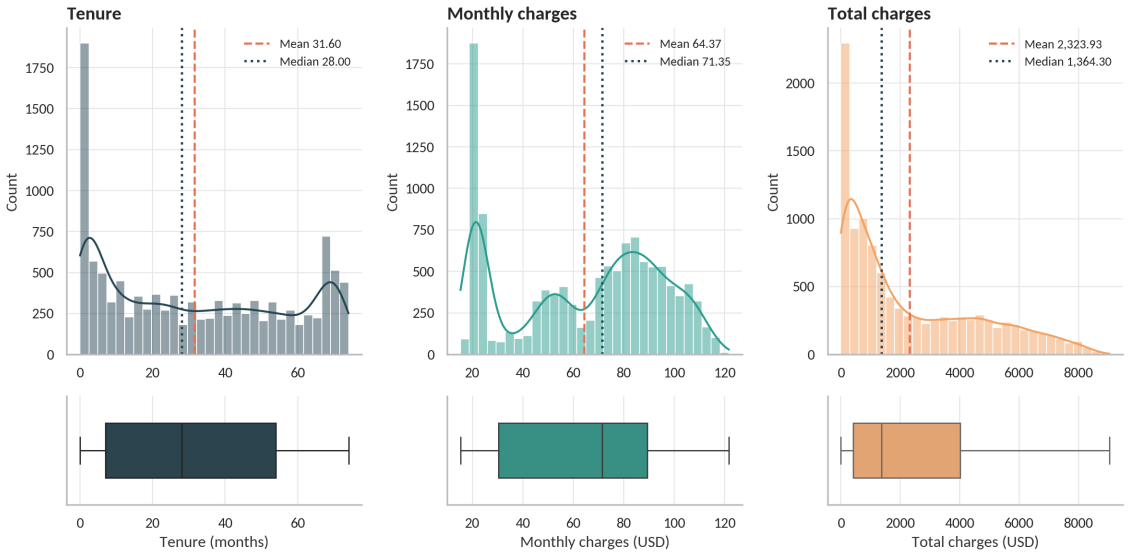


Figure 2. Distribution of tenure, monthly charges and total charges, with box plots below.

Tenure has two clusters: a large group of very new customers (18.50% have three months or less) and another near the 70-month mark (15.01% have 66 months or more), with a fairly even spread between. **Monthly charges** show three humps. The first, around $20, is the 24.28% of customers paying under $30, which are mainly phone-only customers with no internet; the other two sit near $50 and $85 to $100. **Total charges** are skewed to

the right (skewness 0.89) because many customers are recent and have small accumulated bills. The box plots show no points beyond the whiskers.

2.3.3 Categorical variables

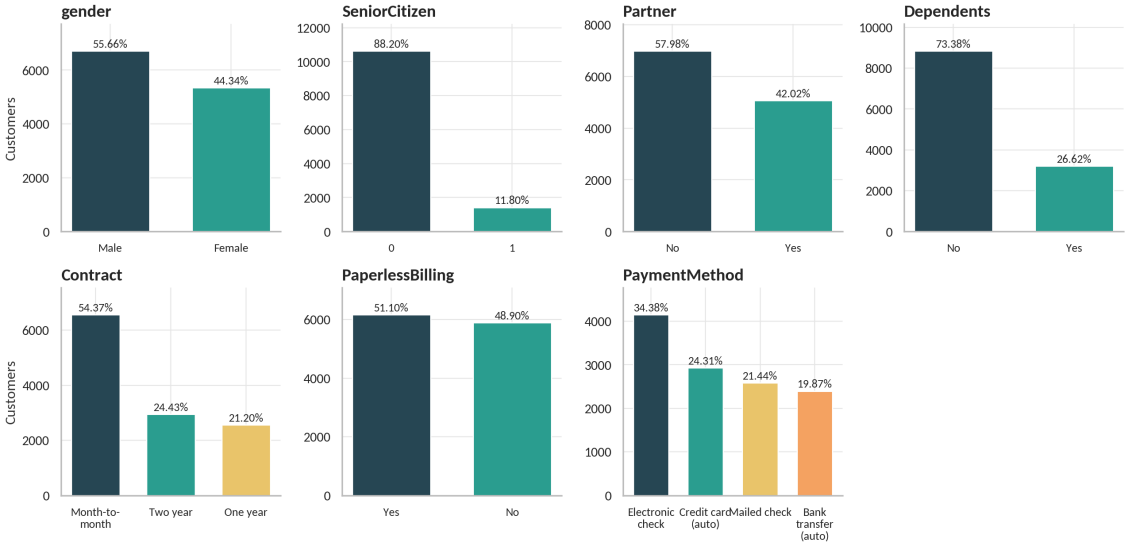


Figure 3. Customer profile and billing columns: number of customers per level.

The base is mostly younger households: only 11.80% are senior citizens, 42.02% have a partner and 26.62% have dependents. 54.37% are on a month-to-month contract, and the most common way to pay is by electronic check (34.38%). Paperless billing is used by 51.10%.

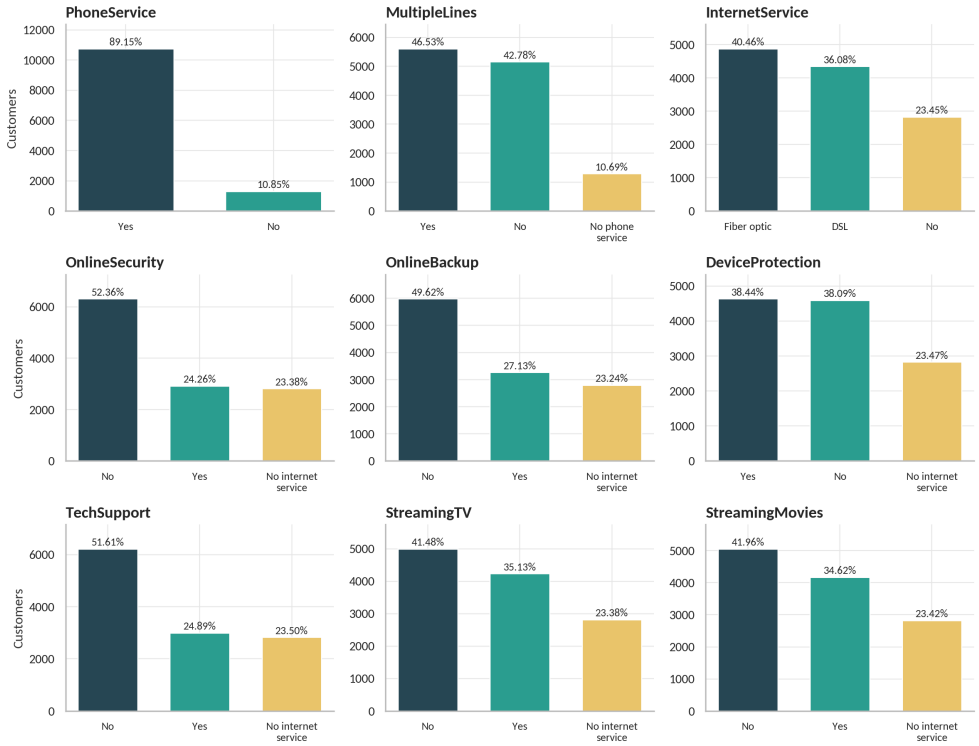


Figure 4. Phone, internet and add-on service columns: number of customers per level.

89.15% have phone service. For internet, fiber optic is the largest group (40.46%), followed by DSL (36.08%), and 23.45% have no internet. That last group is the same share that shows 'No internet service' in all six add-on columns, so those labels repeat what InternetService already says. Even among customers who do have internet, take-up of the protection and support add-ons is low: online security is held by 31.49% of them and tech support by 32.43%.

2.4 Bivariate analysis

2.4.1 Categorical variables against churn

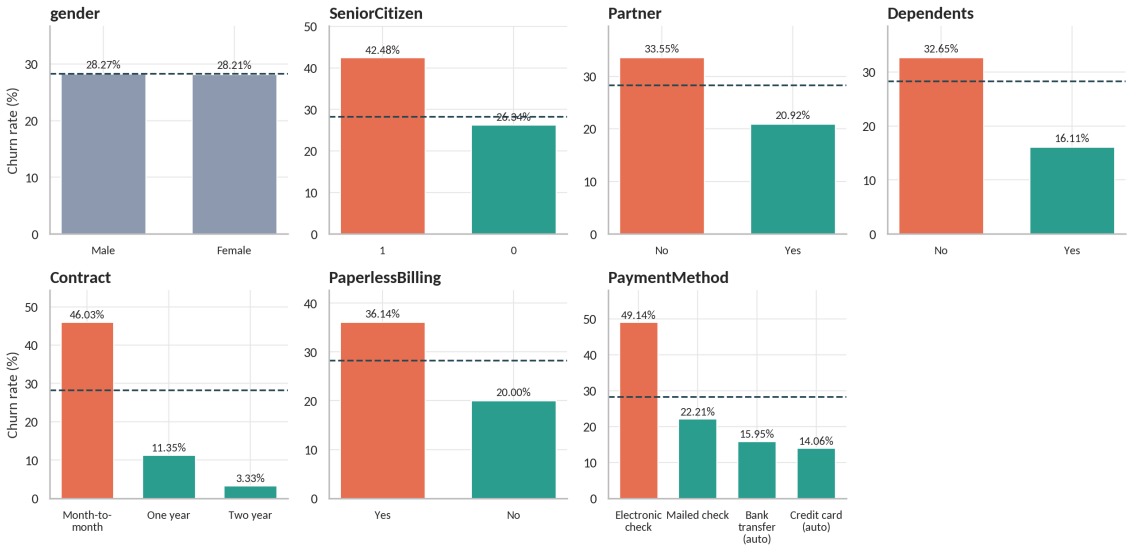


Figure 5. Churn rate by level for the profile and billing columns. The dashed line is the overall rate (28.25%); red bars are more than 5% above it, green more than 5% below.

Contract type gives the widest spread: month-to-month customers churn at 46.03%, against 11.35% on one-year and 3.33% on two-year contracts. Payment method is the next clearest: 49.14% of electronic-check payers leave, while customers on automatic bank transfer (15.95%) or credit card (14.06%) rarely do. Paperless billing (36.14% vs 20.00%), having no partner (33.55% vs 20.92%), having no dependents (32.65% vs 16.11%) and being a senior citizen (42.48% vs 26.34%) all go with higher churn. Gender makes no difference (28.27% for men, 28.21% for women).

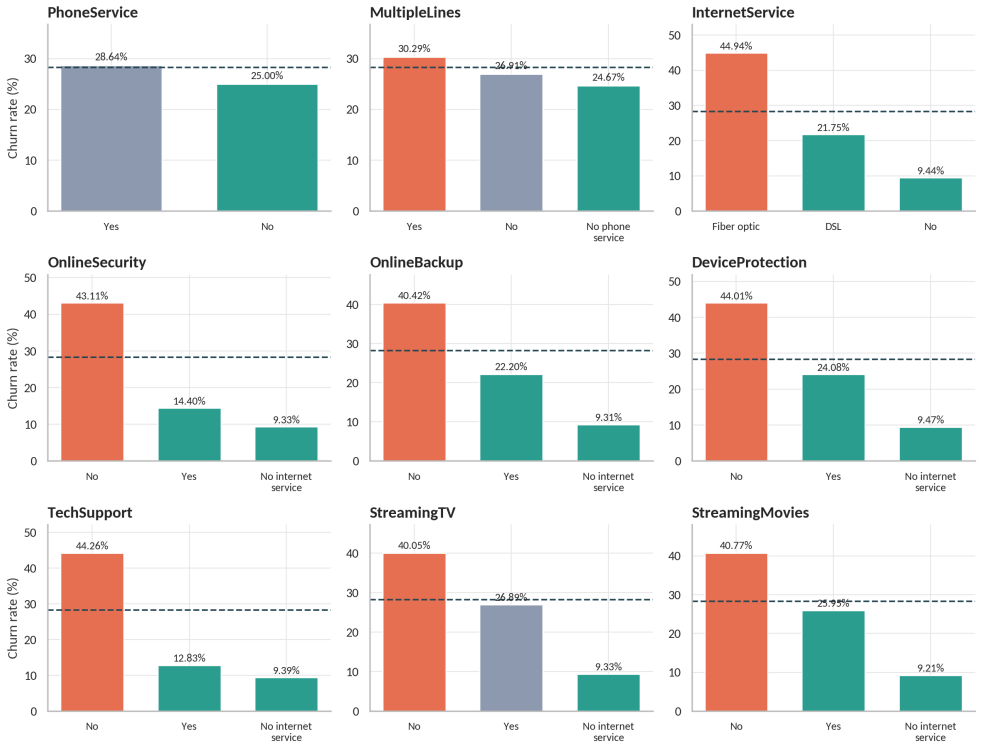


Figure 6. Churn rate by level for the phone, internet and add-on columns.

Fiber optic customers churn at 44.94%, DSL customers at 21.75% and customers without internet at only 9.44%. Within the add-ons, customers who *do not* hold online security churn at 43.11% and those who do at 14.40%; the

same pattern holds for tech support (44.26% vs 12.83%), device protection and online backup. Streaming services and phone-related columns make little difference.

Table 4 puts a number on how strongly each column moves with churn, using a chi-square test and Cramér's V (0 means no link, 1 a perfect link). With more than 12,000 rows almost every test is significant, so the size of V is what ranks the columns.

Table 4. *Association of each categorical column with churn*

Variable	Chi-square	p-value	Cramér's V	Strength
Contract	2,285.53	<0.0001	0.44	Strong
TechSupport	1,636.16	<0.0001	0.37	Strong
OnlineSecurity	1,462.30	<0.0001	0.35	Strong
PaymentMethod	1,409.22	<0.0001	0.34	Strong
InternetService	1,254.19	<0.0001	0.32	Strong
DeviceProtection	1,094.67	<0.0001	0.30	Strong
OnlineBackup	992.10	<0.0001	0.29	Moderate
StreamingMovies	906.88	<0.0001	0.27	Moderate
StreamingTV	845.45	<0.0001	0.26	Moderate
PaperlessBilling	386.20	<0.0001	0.18	Moderate
Dependents	316.90	<0.0001	0.16	Moderate
Partner	230.50	<0.0001	0.14	Weak
SeniorCitizen	160.27	<0.0001	0.12	Weak
MultipleLines	24.21	<0.0001	0.04	Negligible
PhoneService	7.45	0.0064	0.02	Negligible
gender	0.00	0.9602	0.00	Negligible

Contract, tech support, online security, payment method, internet service and device protection are the strongest links. Part of the strength of the add-on columns comes from their 'No internet service' level, which holds the low-churn customers without internet, so they partly stand in for InternetService. Gender and the two phone columns are negligible.

2.4.2 Numeric variables against churn

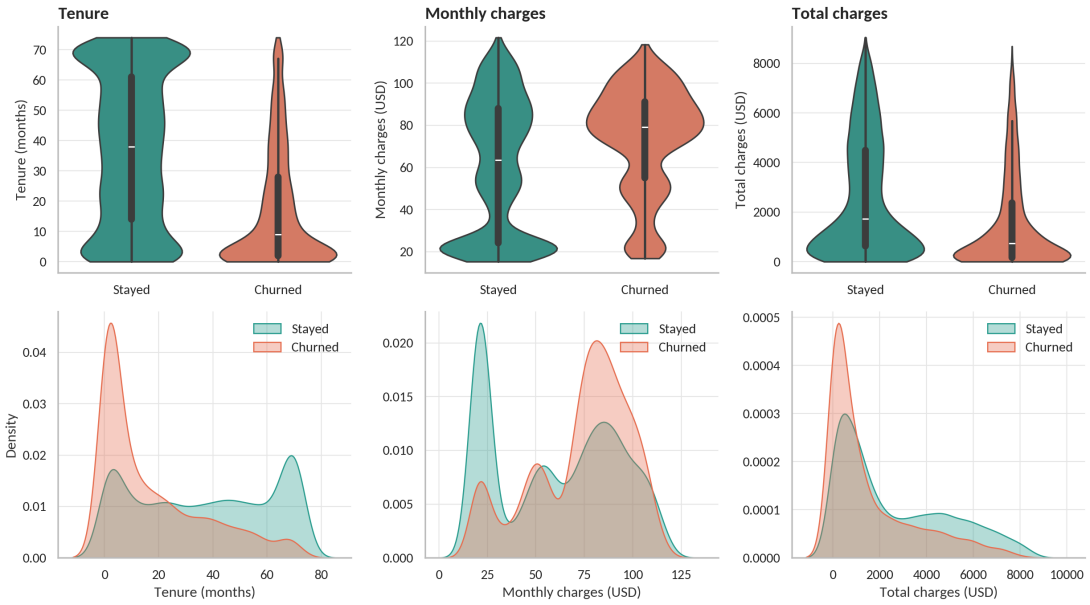


Figure 7. Tenure, monthly charges and total charges for customers who stayed and customers who churned.

Table 5. Numeric variables by churn status

Variable	Mean (stayed)	Mean (churned)	Median (stayed)	Median (churned)	Mann-Whitney p
tenure	37.12	17.34	38.00	9.00	<0.0001
MonthlyCharges	60.79	73.44	63.55	79.25	<0.0001
TotalCharges	2,619.35	1,553.33	1,723.95	747.41	<0.0001

Churners are mostly new customers: their median tenure is 9 months against 38 for stayers. They also pay more each month ($73.44 on average against $60.79). Total charges are lower for churners only because they spent less time as customers. All three differences are statistically clear (Mann-Whitney test, used because the distributions are not bell-shaped).

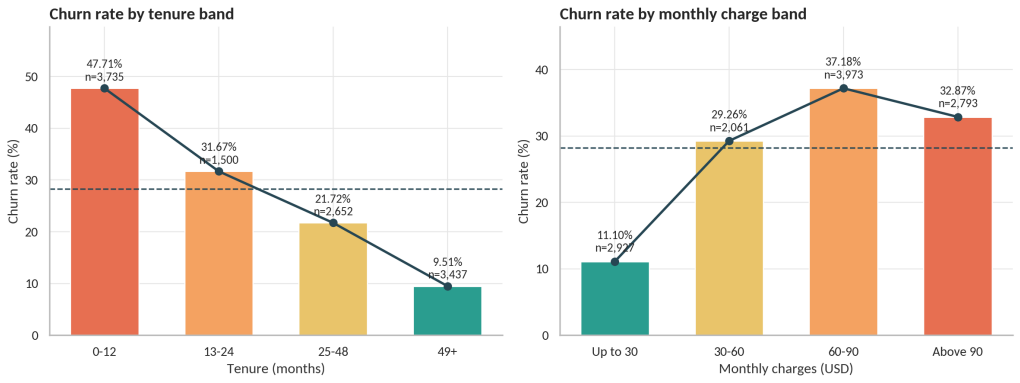


Figure 8. Churn rate by tenure band and by monthly charge band (n = customers in the band).

Churn falls steadily with tenure, from 47.71% in the first year to 9.51% beyond four years. It rises with the size of the bill, from 11.10% for customers paying up to $30 to 32.87% for those paying above $90. The first year is when customers are most at risk.

2.5 Multivariate analysis

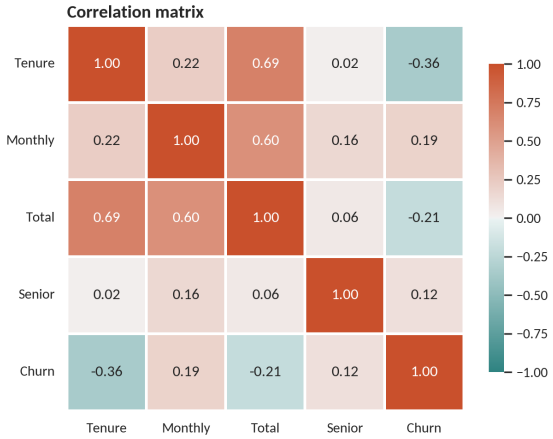


Figure 9. Correlation matrix of the numeric columns and churn.

Tenure and total charges are strongly related (correlation 0.69), as expected because the total is built up month by month. Monthly charges and total charges are related more loosely (0.60). Churn is negatively correlated with tenure (-0.36) and total charges (-0.21), and positively with monthly charges (0.19). The tenure-total overlap is checked again in Section 3.11 because it matters for a linear model.

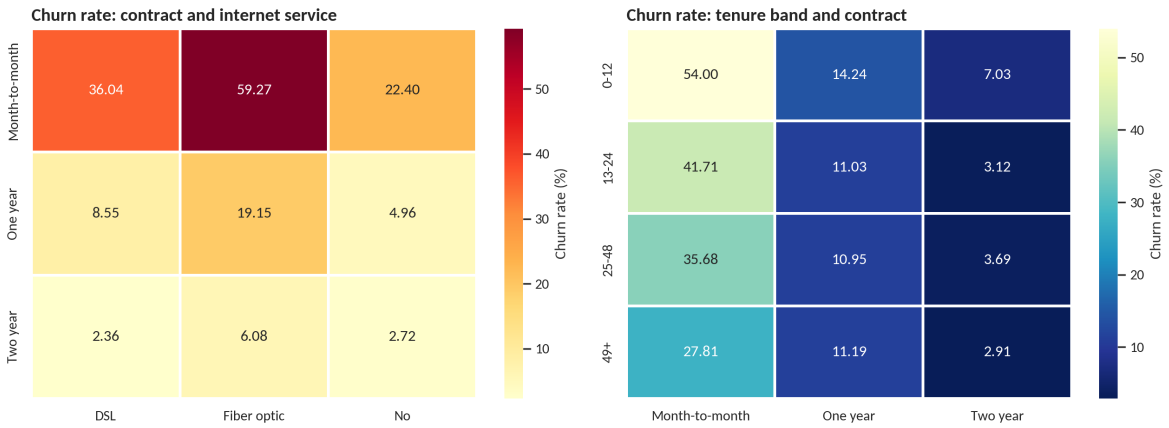


Figure 10. Churn rate (%) for combinations of contract and internet service (left), and of tenure band and contract (right).

Combining columns sharpens the picture. A month-to-month customer on fiber optic churns at 59.27%, while a two-year customer with no internet churns at 2.72%. Contract type protects even within the first year: new customers on a two-year contract churn at 7.03%, against 54.00% for new month-to-month customers. Short tenure and a flexible contract together are the riskiest combination in the data.

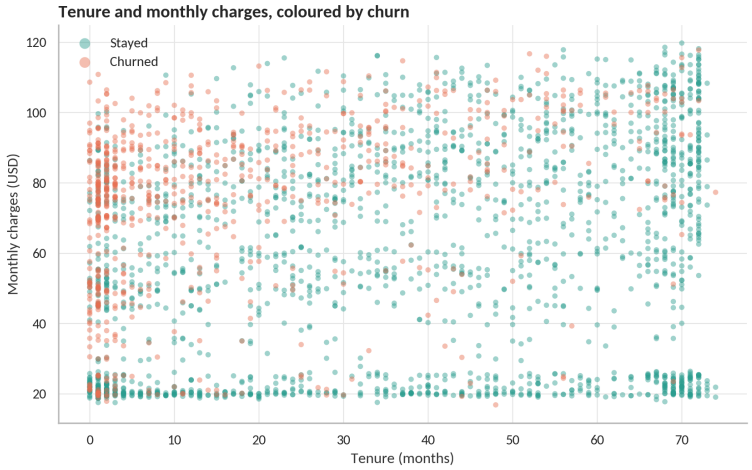


Figure 11. Tenure against monthly charges for a random sample of 3,000 customers, coloured by churn.

Churners (red) cluster in the top-left: short tenure and a high bill. Long-tenure customers are almost all loyal whatever they pay, and customers on the cheapest plans (about $20) hardly ever leave.

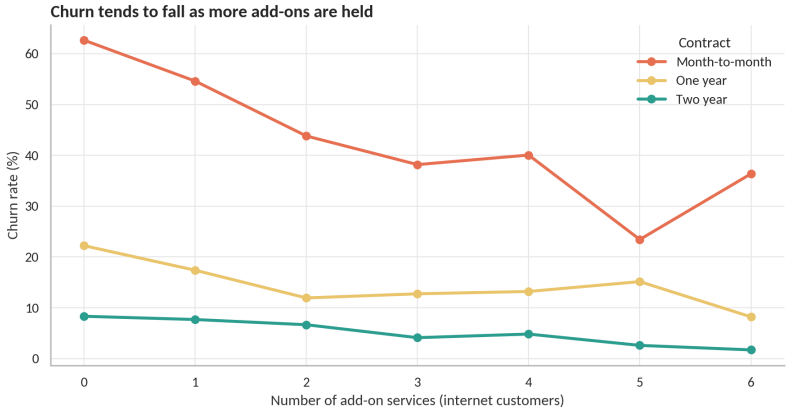


Figure 12. *Churn rate by number of add-on services held, for customers with internet, split by contract type.*

Churn tends to fall as more add-on services are held, within each contract type. Month-to-month customers with no add-on churn at 62.65%, against 36.36% for those with 6, and for two-year customers the rate drops from 8.33% to 1.72%. The line is not perfectly smooth because a few groups are small (only 11 month-to-month customers hold all six). This suggests the services act as a reason to stay, with the caveat that customers who take many add-ons may already be the more committed ones.

2.6 Insights from the EDA

Table 6. *Summary of findings from the exploratory analysis*

What the data shows	Evidence	What it means for AlphaCom
Contract length is the strongest single signal	Churn 46.03% (month-to-month) vs 3.33% (two year); V = 0.44	Move month-to-month customers to longer terms, especially early on
The first year is the danger zone	Churn 47.71% in months 0 to 12, 9.51% after four years	Focus onboarding and early check-ins on new customers
Fiber optic customers leave most	44.94% vs 21.75% for DSL	Look at price, reliability and support for the fiber product
Manual payment goes with leaving	Electronic check 49.14%; automatic methods 14.06% to 15.95%	Encourage a switch to automatic payment
Support and security add-ons go with staying	No tech support 44.26% vs 12.83% with it	Bundle or trial these add-ons for at-risk internet customers
Higher bills go with churn	32.87% churn above $90 a month vs 11.10% up to $30	Check whether expensive plans are seen as good value
Gender and phone columns do not matter	Men 28.27%, women 28.21%	Leave them out of campaign targeting

3. Data Preprocessing

The steps follow the order in which they were carried out. One rule shaped the order: anything that learns from the data (imputation, scaling) is fitted on the training rows only, so the 80:20 split is made before those steps (Section 3.5).

3.1 Duplicate value check

Only 2 rows are identical in the raw file, but once spellings and number formats are standardised 27 rows turn out to repeat an earlier row on all 19 attributes. Without a customer ID the rows cannot be proven to be the same person, and 33% of them are customers with no internet and about one month of tenure, a profile that many genuine customers share. They are 0.22% of the file, so the choice hardly matters; the rows were removed because a repeated record that lands in both the training and test sets would flatter the test score. 12,055 rows became 12,028.

Table 7. Duplicate check

Check	Rows
Exact duplicates in the raw file	2
Identical rows after labels and number formats are standardised	27
Rows removed	27
Rows kept	12,028

3.2 Data cleaning

Two text columns held several spellings of the same value, and the two charge columns had been stored as text. These are fixed row by row, so nothing is learned from the data.

Table 8. Label and type clean-up

Column	Problem in the raw file	Distinct values before	After
Churn	8 spellings of Yes / No	8	2
PaymentMethod	20 spellings of 4 methods	20	4
MonthlyCharges	text such as '$29.85'	-	numeric
TotalCharges	text such as '$1889.5'	-	numeric

3.3 Anomalous value check

Table 9. Anomalies found

Finding	Rows	Treatment
Negative tenure (values -1, -2, -3)	126	Impossible. Set to missing
Negative TotalCharges	147	A cumulative bill cannot be negative. Set to missing
Tenure equal to zero	300	Valid: customers who joined in the current month. Kept
TotalCharges far from tenure × MonthlyCharges	see next table	Kept as reported, tested in Section 3.11
No-internet customers billed above $30 (first 7,043 rows / later rows)	0.00% / 1.92%	Kept; points to the same data-source issue

The data description says TotalCharges need not equal tenure × MonthlyCharges exactly, because of part-month billing, fees and credits. The check below measures how far apart they really are. Such effects should move the ratio by a few percent, not by multiples.

Table 10. Agreement of TotalCharges with tenure × MonthlyCharges

Part of the file	Rows checked	% within 20% of expected	% within 50% of expected	Median ratio
First 7,043 rows	5,853	99.18	99.97	1.00
Remaining rows	3,708	17.72	41.91	1.14
All rows	9,561	67.59	77.45	1.00

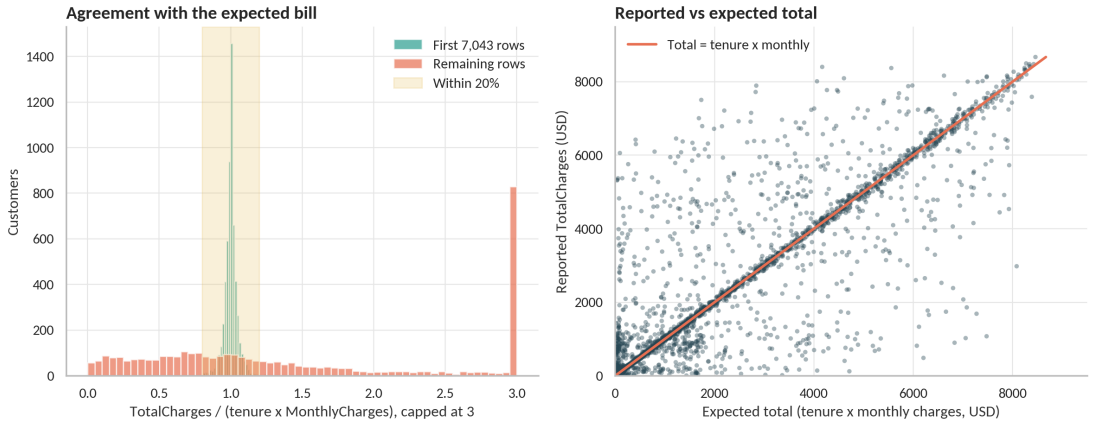


Figure 13. Reported total against the expected total. Left: ratio of the two (values above 3 are shown at 3). Right: the red line is where the two agree.

In the first 7,043 rows the totals follow the expected bill closely (99.18% within 20%). In the remaining rows only 17.72% do, and many totals are several times larger or smaller than they should be. The gap sits in one part of the file, which looks like records merged from another source. The data alone cannot say which of tenure, monthly charges or the total is wrong, so the totals were kept as reported, and Section 3.11 tests whether correcting or dropping them changes the model.

3.4 Missing value check

After the impossible values were blanked, three columns have gaps. 2,231 rows (18.55%) are missing at least one of the three, and most of them are missing just one. All other columns are complete.

Table 11. Missing values by column

Column	Missing	Missing (%)
tenure	728	6.05
MonthlyCharges	301	2.50
TotalCharges	1,351	11.23

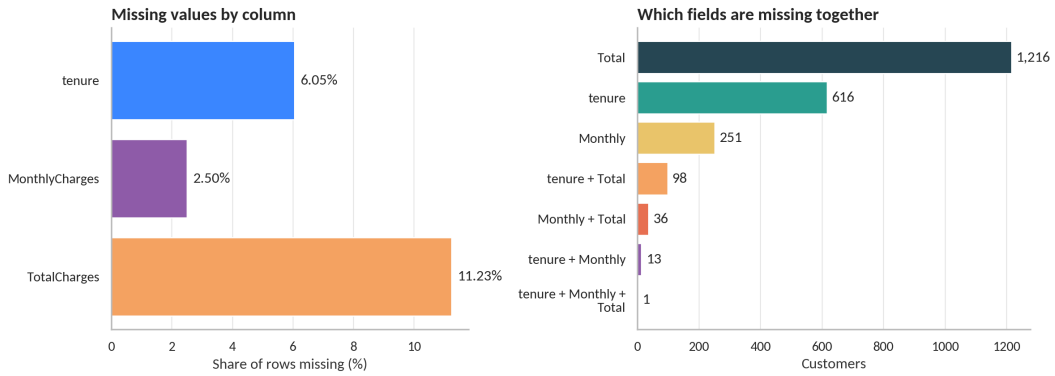


Figure 14. Share of rows missing per column (left) and the combinations in which values are missing (right).

Missingness is not completely random: churn is 33.65% when tenure is missing against 27.87% when it is present, and 32.42% against 27.69% for TotalCharges. The differences are modest, and the gaps can be filled from other columns, so the missing values are imputed rather than the rows dropped.

3.5 Train-test split

The data was split 80:20, stratified on Churn so that both parts keep the same churn share. The test set is held back and used once, for the final scores of the baseline.

Table 12. Train and test sets

Set	Rows	Churned	Churn rate (%)
Train (80%)	9,622	2,715	28.22
Test (20%)	2,406	679	28.22

3.6 Missing value treatment

The categorical columns were first encoded as numbers so that an imputer can use them as clues. The add-on columns were reduced to Yes/No, because 'No internet service' only repeats InternetService.

To choose the method on evidence, 500 known values of each column were hidden in the training set, every method tried to recover them, and the average error was compared (table and chart below).

Table 13. Average error when hidden values are recovered (lower is better)

Method	tenure (months)	MonthlyCharges (USD)	TotalCharges (USD)
Global median	22.11	26.91	1,748.36
Median by group	16.19	11.19	1,623.27
Billing identity	10.53	57.10	580.13
Iterative (linear)	11.71	6.55	898.56
Iterative (random forest)	9.04	6.47	631.44

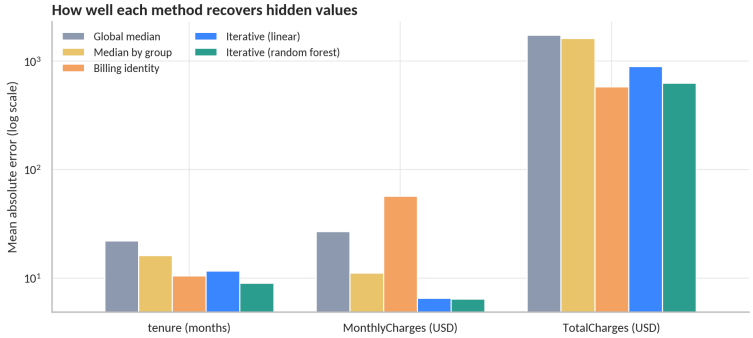


Figure 15. Average error of each imputation method (log scale).

A random forest that predicts each missing column from all the others (iterative imputation) is best for tenure and for monthly charges. For TotalCharges the simple rule *tenure × MonthlyCharges* is best, with an error of $580.13 against $631.44. The final treatment therefore uses the forest for tenure and monthly charges (tenure rounded to whole months, both kept within the observed range) and the rule for missing totals. The imputer was fitted on the training set and then applied to both sets, leaving no missing values (checked in the notebook). The filled-in distributions follow the observed ones closely (see the additional figures at the end of the report).

3.7 Outlier check and treatment

Table 14. *Outlier check with the interquartile range (IQR) rule (fences at 1.5 × IQR beyond the quartiles)*

Variable	Q1	Q3	Lower fence	Upper fence	Min	Max	Outside fences	|z| > 3
tenure	7.00	54.00	-63.50	124.50	0.00	74.00	0	0
MonthlyCharges	30.40	89.38	-58.07	177.85	15.29	121.67	0	0
TotalCharges	422.60	4,011.69	-4,961.03	9,395.33	0.00	9,039.92	0	0

No value lies outside the fences and none is more than three standard deviations from the mean, so no outlier treatment was needed. The largest totals belong to customers with long tenure on expensive plans, which are real values the model should see.

3.8 Feature engineering

Four features were built from existing columns: the natural log of tenure, the number of add-on services, an autopay flag and a household flag (partner or dependents). Each idea was tested by 5-fold cross-validation on the training set with logistic regression, since that is the baseline model. Four decimals are shown because the differences are small.

Table 15. *Cross-validated results for the feature sets*

Feature set	Recall	F1	ROC-AUC
A. Original encoded columns	0.8000	0.6511	0.8493
B. A + log(tenure)	0.8029	0.6555	0.8526
C. A with tenure replaced by log(tenure)	0.8029	0.6542	0.8526
D. A + AutoPay + HasFamily	0.7989	0.6507	0.8493
E. A with six add-ons replaced by their count	0.7971	0.6495	0.8470

Adding the log of tenure (variant B) or using it in place of tenure (variant C) gives the best ROC-AUC, because churn falls quickly over the first months and then levels off, which a straight line in tenure cannot follow. The two are practically tied, so C was chosen: it has one column fewer and does not carry tenure twice. The other ideas add nothing for a linear model since they only repeat information already in the columns; the add-on count is even slightly worse because it hides which add-on matters. They remain in the prepared data for the tree-based models planned for the next stage. The baseline uses variant C, with 23 features.

3.9 Class imbalance

Table 16. Class counts

Class	Train	Test
Stayed	6,907	1,727
Churned	2,715	679

Churners are about 28% of customers, a ratio of 2.54 to 1. That is enough for a model to look good by favouring 'stayed', but not extreme enough to need artificial customers. The imbalance is handled with balanced class weights, which make each churner count about 2.5 times as much in the loss. This adds no invented rows and leaves the test set untouched. The split itself is stratified. Section 4.3 shows what the weights change.

3.10 Feature scaling and data preparation for modelling

Logistic regression is sensitive to the size of its inputs, so the three continuous columns (log tenure, monthly charges, total charges) were standardised to mean 0 and standard deviation (SD) 1, using the training set only. The other columns are 0/1 indicators and stay as they are. Categorical columns became indicators with a reference group: no internet, month-to-month contract and electronic check, so that coefficients read as a comparison with those groups. The prepared matrices have 23 columns, 9,622 training rows and 2,406 test rows, with no missing values.

Table 17. Scaled columns: the training set is exactly standardised; the test set is close

Column	Train mean	Train SD	Test mean	Test SD
log_tenure	0.00	1.00	0.02	0.99
MonthlyCharges	0.00	1.00	-0.02	0.99
TotalCharges	0.00	1.00	0.00	0.98

3.11 Multicollinearity and the TotalCharges choice

Variance inflation factors show how much each column is explained by the others; values above 10 are normally a concern.

Table 18. Five highest variance inflation factors

Column	VIF
InternetService_Fiber optic	10.18
MonthlyCharges	9.00
InternetService_DSL	5.86
TotalCharges	3.34
Contract_Two year	2.15

Fiber optic (10.18) and monthly charges (9.00) are high because the price of a plan is largely set by the internet type. Both are real drivers, so they stay; the regularised model copes with the overlap. TotalCharges, despite its link with tenure, is acceptable at 3.34. To check that the awkward totals do not drive the results, the whole pipeline was repeated with the totals as reported, with totals more than 50% away from the expected bill replaced by that expected bill, and with the column dropped.

Table 19. Test-set results under three treatments of TotalCharges

Version	Recall	Precision	F1	ROC-AUC
(a) As reported	0.7909	0.5756	0.6663	0.8563
(b) Inconsistent totals corrected	0.7894	0.5702	0.6621	0.8550
(c) TotalCharges dropped	0.7953	0.5690	0.6634	0.8545

The three versions are almost identical, so the treatment of TotalCharges does not drive the model. The as-reported version is kept because it uses the data as recorded.

3.12 Data leakage review

Table 20. Leakage checks

Risk	How it was handled
Test data influencing preprocessing	Split first. The imputer and the scaler were fitted on the training set only.
Target used to fill or build features	Churn was never an input to the imputer or to any engineered feature.
A column that already contains the outcome	No column records a cancellation date, reason or final bill. TotalCharges is a running total of past billing.
The same record in both sets	Repeated rows were removed before the split.
Choosing settings on the test set	Feature choices were made by cross-validation on the training set. The test set is used once, for the baseline scores.

4. Model Building: Baseline Model

4.1 Metric of choice

Two kinds of mistake are possible. A **missed churner** (false negative) loses the customer and every future month of that customer's bill. A **false alarm** (false positive) costs one retention offer to someone who would have stayed. The first mistake is normally much more expensive than the second, so the model is judged mainly on **recall for the churn class**: the share of real churners it finds.

Recall alone can be gamed by flagging every customer, so precision and F1 are watched alongside it. ROC-AUC and the precision-recall area show how well the model ranks customers whatever cut-off is used. Plain accuracy is not used for selection: predicting that nobody churns would already be 72% accurate while catching no one.

4.2 Baseline model

Logistic regression was chosen as the baseline because it is quick, needs little tuning and gives coefficients that business teams can read. It was trained on the 23 prepared features with balanced class weights. Two reference points were added: a dummy model that always predicts the majority class, and the same logistic regression without class weights.

Table 21. Baseline and reference models on the test set (2,406 customers)

Model (test set)	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC
Majority-class dummy	0.72	0.00	0.00	0.00	0.50	0.28
Logistic regression (no weights)	0.81	0.70	0.55	0.62	0.86	0.70
Logistic regression (balanced weights)	0.78	0.58	0.79	0.67	0.86	0.70

The dummy model reaches 0.72 accuracy and zero recall, which is why accuracy is not a useful yardstick here. Without class weights, logistic regression is more accurate (0.81) and more precise (0.70), but finds only 0.55 of the churners. With balanced weights recall rises to 0.79 while precision falls to 0.58. The ROC-AUC is the same (0.86) for both, which shows that the weights do not change how well customers are ranked; they move the cut-off towards catching more churners. Since missed churners are the costly mistake, the weighted version is the baseline.

4.3 Performance of the baseline

Table 22. Baseline performance on training data, test data and cross-validation

Metric	Train	Test	5-fold CV mean	CV SD
Accuracy	0.76	0.78	0.76	0.01
Precision	0.55	0.58	0.55	0.01
Recall	0.80	0.79	0.80	0.01
F1	0.66	0.67	0.65	0.01
ROC-AUC	0.85	0.86	0.85	0.01
PR-AUC	0.69	0.70	-	-

Training and test scores are close (recall 0.80 and 0.79; ROC-AUC 0.85 and 0.86), and the cross-validation scores vary by only about 0.01 between folds. The model is therefore stable and not over-fitted, though it is a simple linear model and likely under-fits some patterns.

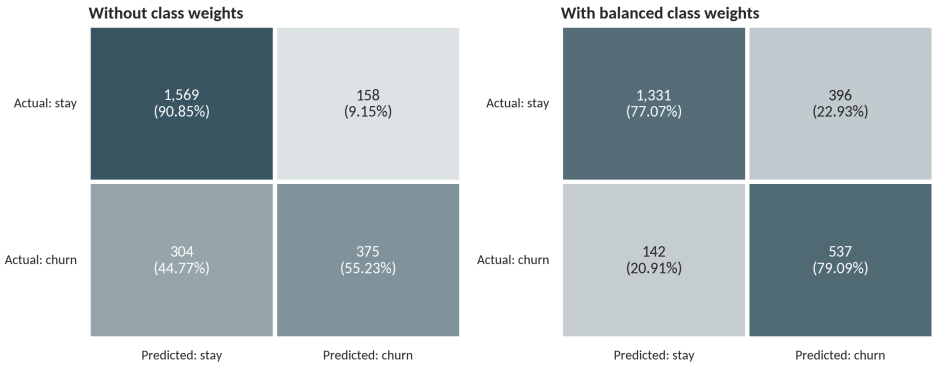


Figure 16. Confusion matrices on the test set, as counts and as a share of each actual class.

With balanced weights the model finds 537 of the 679 customers who churned (recall 0.79) and misses 142. Of the 933 customers it flags, 537 really churn (precision 0.58) and 396 are false alarms. Compared with the unweighted model it catches 162 more churners at the price of 238 more false alarms, about 1.5 extra false alarms for every extra churner found.

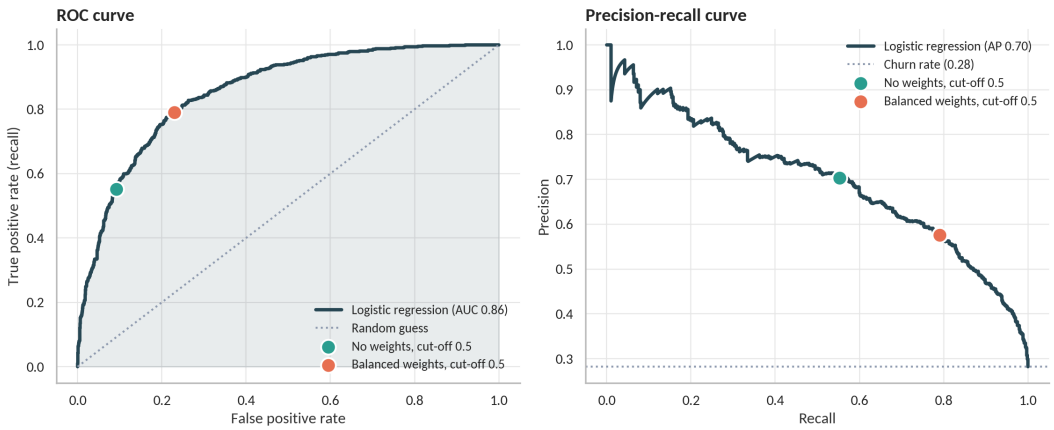


Figure 17. ROC curve (left) and precision-recall curve (right) for the baseline. The dots mark the two versions at the default 0.5 cut-off.

Both curves sit well away from the chance lines, with ROC-AUC of 0.86. The two dots lie on the same curve: class weights do not make the model better at ranking, they choose a point further along it, with higher recall and lower precision. A retention team can choose the point that suits its budget.

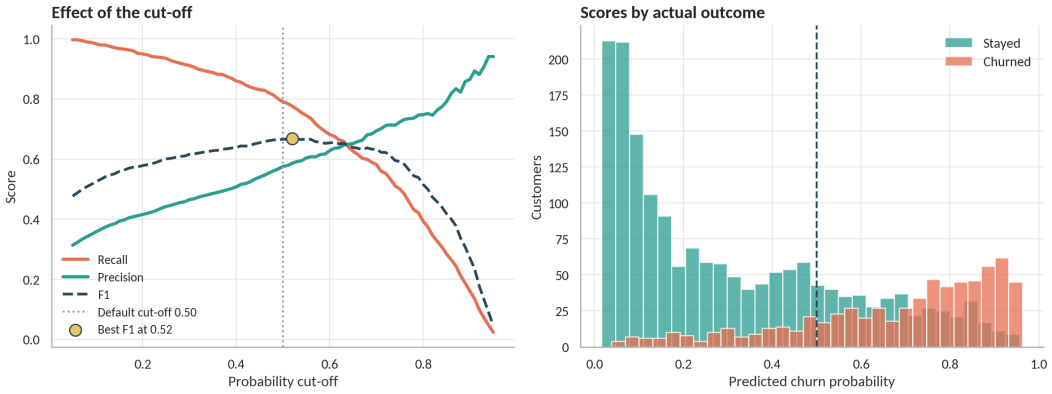


Figure 18. Recall, precision and F1 at different cut-offs (left) and the predicted churn probability of customers who stayed and customers who churned (right).

Raising the cut-off trades recall for precision. F1 is highest at a cut-off of 0.52, almost the default 0.50, so there is no gain from moving it if the goal is balance; a team with a small budget could use a higher cut-off, and one that wants to miss as few churners as possible a lower one. The histogram shows the two groups separate clearly: stayers pile up near zero, churners lean towards one.

4.4 What drives churn in the model

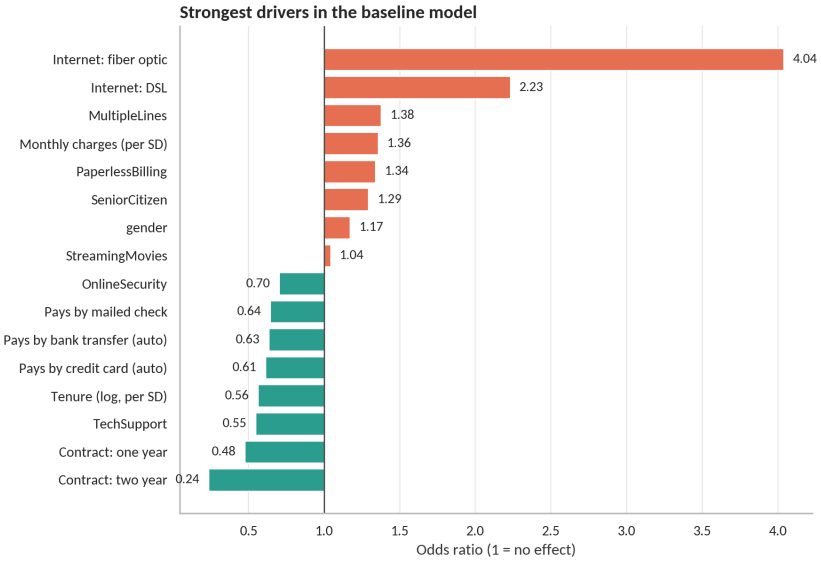


Figure 19. Odds ratios of the strongest drivers. Red raises churn risk, green lowers it. Internet, contract and payment effects are measured against no internet, month-to-month and electronic check.

The model confirms the EDA. Customers with fiber optic (4.04) or DSL (2.23) have higher odds of leaving than those with no internet. A two-year contract cuts the odds to 0.24 times those of a month-to-month customer, and a one-year contract to 0.48. Tech support (0.55) and online security (0.70) lower the odds, and so does paying automatically instead of by electronic check (about 0.61 to 0.63). Longer tenure lowers the odds (0.56 per standard deviation of log tenure), while higher monthly charges (1.36 per SD), paperless billing (1.34), multiple lines (1.38) and being a senior citizen (1.29) raise them. Gender shows a small coefficient (1.17) even though raw churn rates for men and women are equal, and it is not treated as a business finding.

4.5 How useful is the ranking for retention?

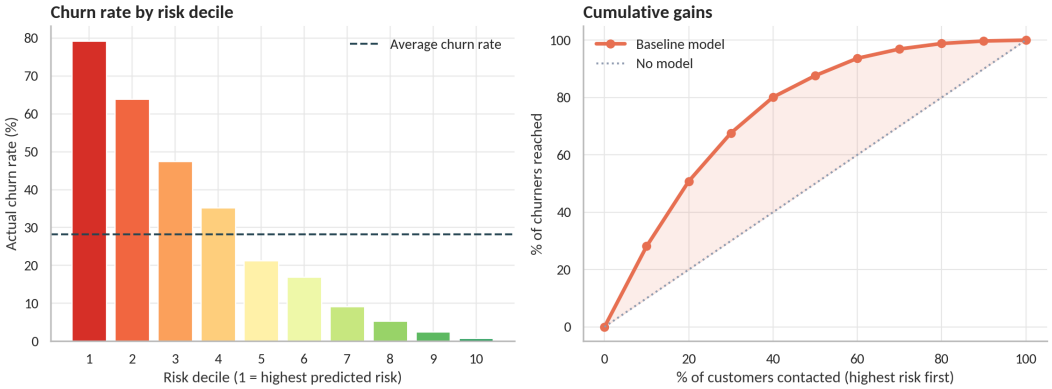


Figure 20. Actual churn rate in each risk decile of the test set (left) and the cumulative share of churners reached (right).

Sorting customers by predicted risk concentrates the churners well. In the top decile 79.25% of customers actually churned, 2.81 times the average rate; in the bottom decile only 0.83%. Contacting the highest-risk 20% of customers would reach 50.81% of all churners, and the top 40% would reach 80.12%. A campaign aimed at the top two deciles would reach about 2.5 times as many churners per call as a random mailing.

Table 23. Gains table for the test set

Risk decile	Customers	Churners	Churn rate (%)	Cumulative % of churners	Lift
1	241	191	79.25	28.13	2.81
2	241	154	63.90	50.81	2.26
3	240	114	47.50	67.60	1.68
4	241	85	35.27	80.12	1.25
5	240	51	21.25	87.63	0.75
6	241	41	17.01	93.67	0.60
7	240	22	9.17	96.91	0.32
8	241	13	5.39	98.82	0.19
9	240	6	2.50	99.71	0.09
10	241	2	0.83	100.00	0.03

4.6 Summary and next step

The baseline logistic regression finds 0.79 of the churners with a precision of 0.58, an F1 of 0.67 and a ROC-AUC of 0.86. It is stable between training, test and cross-validation and its drivers match what the exploration showed. Its main limits are that it is linear, so it cannot capture interactions such as short tenure combined with a flexible contract, and that about 42% of the customers it flags are false alarms. The advanced models in the final submission will be judged against these figures, mainly on recall, with F1 and ROC-AUC alongside.

Additional figures

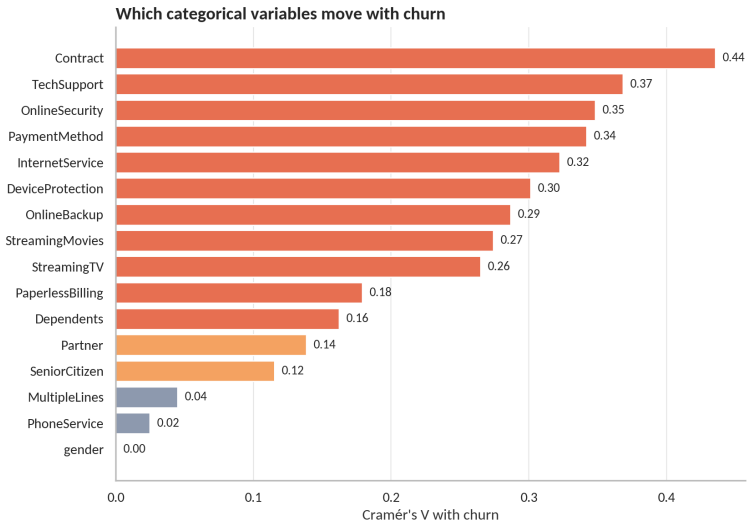


Figure 21. *Cramér's V of each categorical column with churn (values in Table 4).*

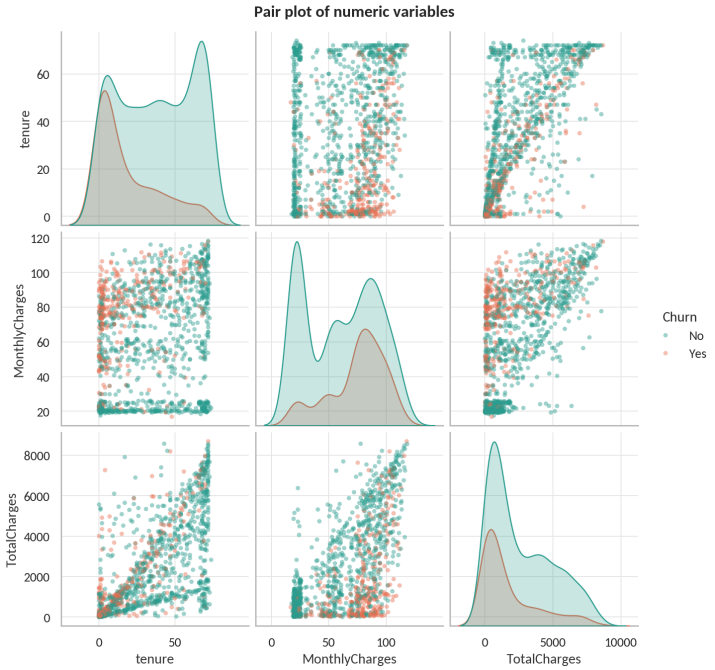


Figure 22. *Pair plot of tenure, monthly charges and total charges for 1,500 random customers.*

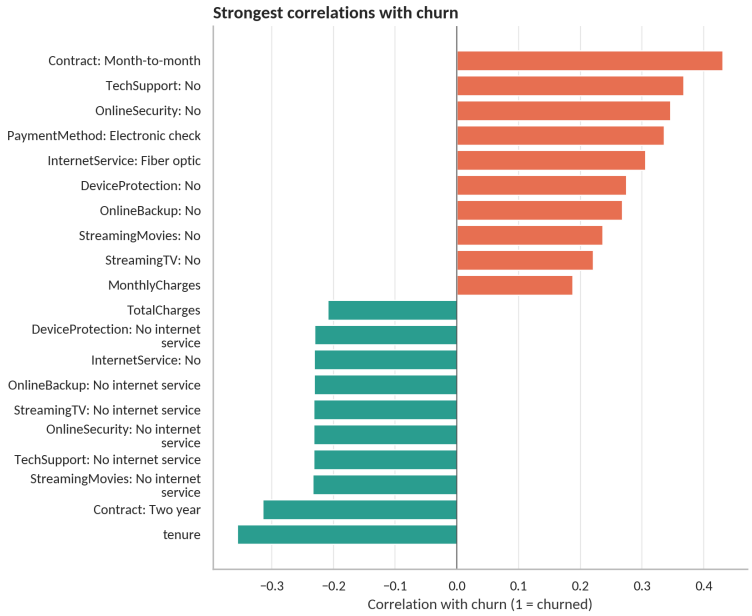


Figure 23. *The ten strongest negative and positive correlations of any (encoded) column with churn.*

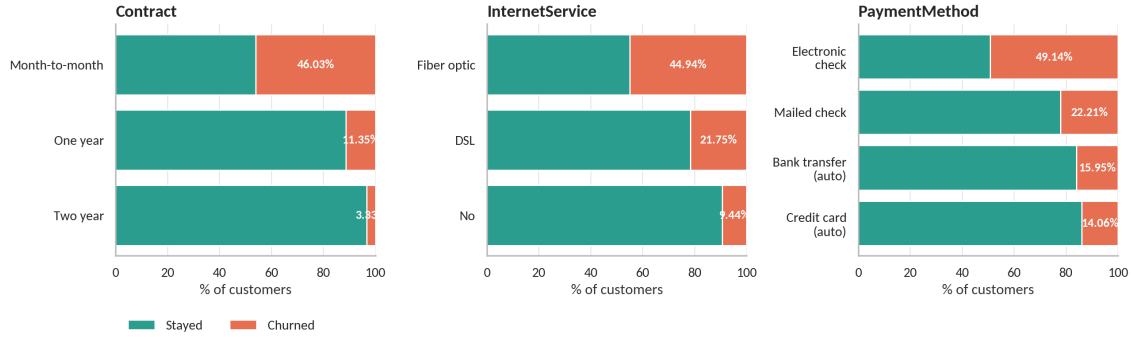


Figure 24. *Share of customers who stayed and churned within the main driver columns.*

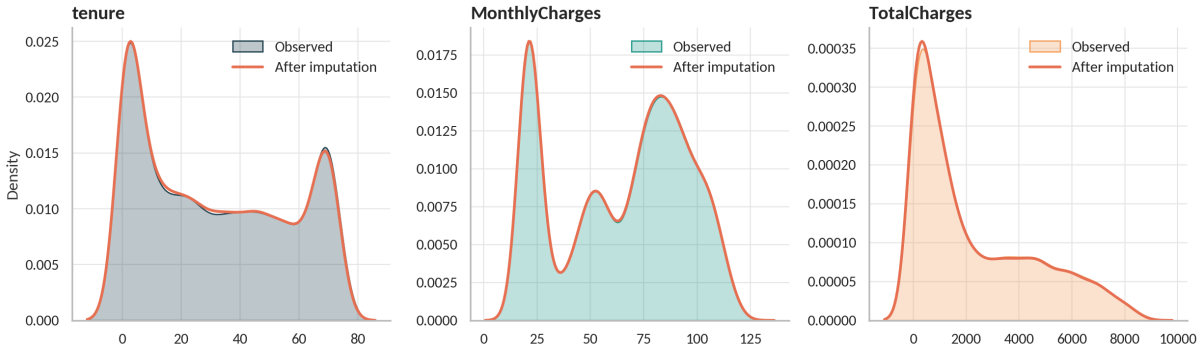


Figure 25. *Distribution of observed values against the values after imputation (training set).*

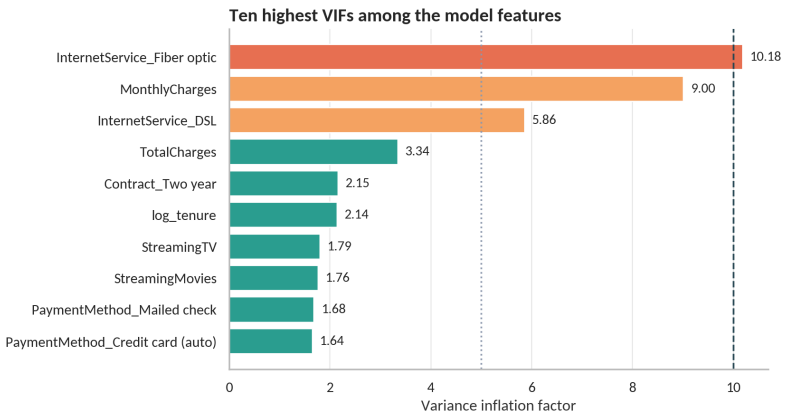


Figure 26. *Ten highest variance inflation factors among the model features.*